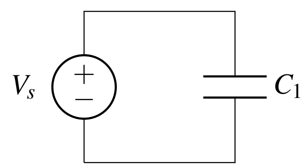
EECS 16A Designing Information Devices and Systems I
Discussion 10A

1. Voltages Across Capacitors

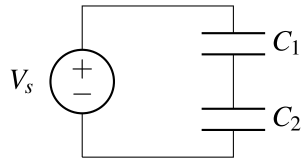
For the circuits given below, determine the voltage across each capacitor and calculate the charge and energy stored on each capacitor (assume all capacitors start *uncharged*, and then we've let the system reach steady state). We are also given $C_1 = 1\,\mu\text{F}$, $C_2 = 3\,\mu\text{F}$, and $V_s = 1\,\text{V}$.

Recall charge has units of Coulombs (C), and capacitance is measured in Farads (F) $= \frac{\text{Coulomb}}{\text{Volt}}$.
It may also help to note metric prefix examples: $3\,\mu\text{F} = 3 \times 10^{-6}\text{F}$.

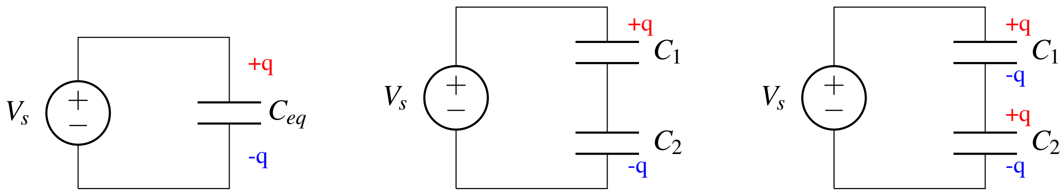
(a)



(b)



Helpful diagrams for considering the charges capacitors linked in series:
(without any initial charges)



Left: Our series capacitors may be modeled as one equivalent capacitor C_{eq}, which after some time is charged up by V_s to have $+q$ on the top plate and $-q$ on the bottom plate.

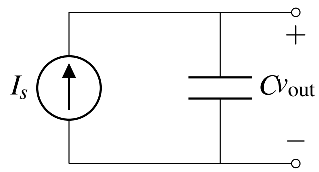
Middle: We return to the 2-capacitor picture, but carry this insight of equivalent charge with us. Now the charge $+q$ is on the top plate *of capacitor* C_1, and $-q$ is on the bottom plate *of capacitor* C_2.

Right: Since capacitor plates have opposite & equal charges, we attain this final right digram.

As another conceptual check, we notice that the node between C_1 and C_2 is isolated from any other connections and should always remain *charge neutral*. From the diagram right we see this is maintained since $(+q)+(-q)=0$.

2. Current Sources And Capacitors

Given the circuit below, find an expression for $v_{\text{out}}(t)$ in terms of I_s, C, V_0, and t, where V_0 is the initial voltage across the capacitor at $t = 0$.



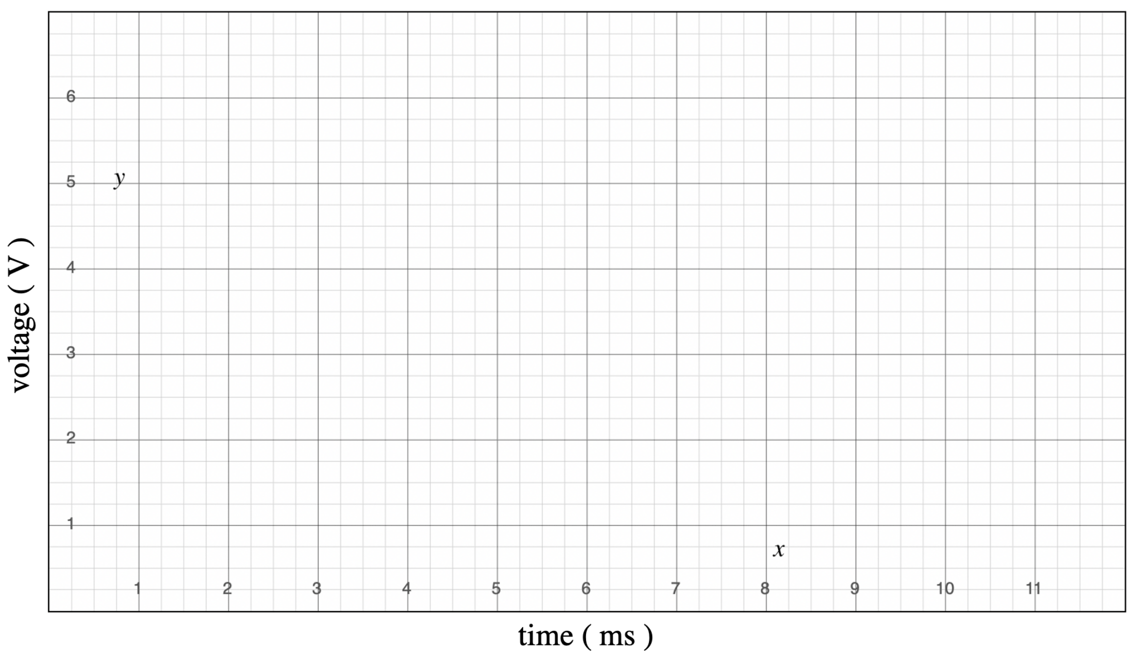
Then plot the function $v_{\text{out}}(t)$ over time on the graph below for the following conditions detailed below. Use the values $I_s = 1\text{mA}$ and $C = 2\,\mu\text{F}$.

(a) Capacitor is initially uncharged $V_0 = 0$ at $t = 0$.

(b) Capacitor has been charged with $V_0 = +1.5\text{V}$ at $t = 0$.

(c) **Practice:** Swap this capacitor for one with half the capacitance $C = 1\,\mu\text{F}$, which is initially uncharged $V_0 = 0$ at $t = 0$.

HINT: Recall the calculus identity $\int_a^b f'(x)dx = f(b) - f(a)$, where $f'(x) = \frac{df}{dx}$.

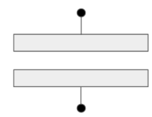


3. Additional reading: capacitance and equivalence

This serves as an aid for students to better master parallel plate capacitors and their equivalence formulae.

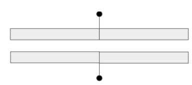
In this course we discuss *parallel-plate* capacitors, which are two closely-spaced metal plates. Additionally we provide the equation for its capacitance:

$$C = \varepsilon \frac{A}{d}$$



Here the plates have an area of $A = L \times W$ (where L is length going into the page and W is the width) and are spaced a distance d apart.

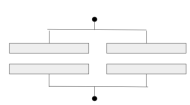
(a) Let's consider a capacitor with the same L and d dimensions as the capacitor described above, yet with twice the width $2W$. This is illustrated below



Since the capacitor plates have been doubled in area, the new capacitance is

$$C = \varepsilon \frac{(2W) \times L}{d} = 2\left(\varepsilon \frac{A}{d}\right).$$

(b) Let's next consider two capacitors each with identical dimensions as in the first example (W, L, d) arranged in parallel as shown below. With the exercise above in mind, what do you think the capacitance of this circuit is?



Intuitively, this should agree with our expression found above, since both systems appear as one capacitor with twice the plate area. Thus, the answer remains $C_{eq} = 2\left(\varepsilon \frac{A}{d}\right)$**.**